

Curriculum and Syllabus



MASTER OF COMPUTER APPLICATION

(MCA- 2 years)

(With Effect from 2020 - 21)

Department of Computer Science and Application

Mahatma Gandhi Kashi Vidyapith, Varanasi

221002

Uttar Pradesh

Master of Computer Application (Two Years)

M.C.A. Ist Semester (w.e.f. 2020-21)

Paper Code	Name of Paper	L	T	P	C	Marks		
						External	Internal	Total
MCA01001	Fundamentals of Computers	6	0	0	6	75	25	100
MCA01002	Introduction to Digital Logic	6	0	0	6	75	25	100
MCA01003	Business Data Processing	6	0	0	6	75	25	100
MCA01004	Design Methods and Analysis of Algorithms	6	0	0	6	75	25	100
MCA01005	PC Software Tools	6	0	0	6	75	25	100
MCA01006	Practical Examination (Based on Theory Papers)	0	0	3	3			100
	LABORATORY 1(DAA LAB)	0	0	3	3			100
	LABORATORY 2 (PC SOFTWARE LAB)							
	Total				36			700

Master of Computer Application (MCA)
M.C.A. IIND Semester (w.e.f. 2020-21)

Paper Code	Name of Paper	L	T	P	C	Marks		
						External	Internal	Total
MCA02001	C Programming	6	0	0	6	75	25	100
MCA02002	Database Management Systems	6	0	0	6	75	25	100
MCA02003	Web Technology and Information Retrieval	6	0	0	6	75	25	100
MCA02004	Cyber Security and Cyber Laws	6	0	0	6	75	25	100
MCA02005	System Programming	6	0	0	6	75	25	100
MCA02006	Practical Examination (Based on Theory Papers)	0	0	3	3			100
	LABORATORY 1(C Programming) LABORATORY 2 (Database Management Systems)	0	0	3	3			100
	Total				36			700

Master of Computer Application (MCA)
Third Semester Syllabus (w.e.f. 2020-21)

Paper Code	Name of Paper	L	T	P	C	Marks		
						External	Internal	Total
MCA03001	Operating Systems	6	0	0	6	75	25	100
MCA03002	Discrete Mathematical Structures	6	0	0	6	75	25	100
MCA03003	Graph Theory and Combinatorics	6	0	0	6	75	25	100
MCA03004	Theory Of Computation	6	0	0	6	75	25	100
MCA03005	Object oriented Programming System with JAVA	6	0	0	6	75	25	100
MCA03006	Practical Examination (Based on Theory Papers)	0	0	3	3			100
	LABORATORY 1 (Operating Systems DOS / UNIX / WINDOWS) LABORATORY 2 (Object oriented Programming System with JAVA)	0	0	3	3			100
	Total				36			700

Master of Computer Application (MCA)
Fourth Semester Syllabus (w.e.f. 2020-21)

Paper Code	Name of Paper	L	T	P	C	Marks		
						External	Internal	Total
MCA04001	Software Engineering	6	0	0	6	75	25	100
MCA04002	Artificial Intelligence	6	0	0	6	75	25	100
MCA04003	Computer Communication Networks	6	0	0	6	75	25	100
MCA04004* (Departmental Elective)	Elective Paper (Any one of the following) MCA04004A: Computer Graphics MCA04004B: Digital Image Processing MCA04004C: Machine Learning	6	0	0	6	75	25	100
MCA04005* (Departmental Elective)	Elective Paper (Any one of the following) MCA04005A: Soft Computing MCA04005B: Data Warehousing And Data Mining MCA04005C: Software Project Management	6	0	0	6	75	25	100
MCA04006A	Practical Examination (Based on Theory Papers) LABORATORY 1(AI and PROLOG)	0	0	3	3			100
MCA04006B	Major Project	0	0	3	3			100
	Total				36			700

PROGRAM OUTCOMES (POs)

POs describe the expected knowledge of students or their ability by the time of completion of post-graduation from the program. The Program Outcomes of the MCA Program are described as below:

PO-1: Strong foundation in core Computer Science and Applications, both theoretical and applied concepts.

PO-2: The ability to apply knowledge of mathematics, science, and engineering to real life problem solving.

PO-3: The ability to analyze, design, model, and develop complex software and information systems.

PO-4: The ability to function effectively within teams.

PO-5: To develop an understanding of professional and ethical responsibility.

PO-6: The ability to communicate effectively, in both written and Oral Communication.

PO-7: To develop an understanding of the impact of information systems solutions in the societal and human context.

PO-8: The ability to engage in lifelong learning.

PO-9: To acquire an up to date Knowledge of contemporary issues and recent Advancements in the field, especially thrust areas such as AI and Machine Learning.

PROGRAM SPECIFIC OUTCOMES (PSOs)

On completion of the Master of Computer Applications degree, the graduates will be able to:

PSO-1: Design, develop and implement interdisciplinary application software projects for efficient design of computer-based systems of varying complexity, in order to meet the demands of industry requirements using modern tools and technologies.

PSO-2: It is intended to provide Industry oriented education in applied computer science and to pursue careers in IT industry/ consultancy/ research and development, teaching and allied areas related to Computer Science.

Program Name: M.C.A. (Master in Computer Applications)

System: CBCS/Elective/Non Elective

Mapping of Program by Courses covering curriculum matrix related to NAAC Criterion

Related Matrix	Sem- I (Course Code)	Sem- II(Course Code)	Sem- III (Course Code)	Sem- IV (Course Code)	Sem-V (Course Code)	Sem- VI (Course Code)
1.1.1 Content Covering Global, National, Local Development Need	MCA01005(All Units)	MCA02004(All Units)	MCA01005(All Units)	MCA04002(All Units) MCA04004C(All Units)		
1.1.3 courses/papers having focus on employability/ Entrepreneurship/ skill development	MCA01001(All Units) MCA01005(All Units)	MCA02001(All Units) MCA02002(All Units) MCA02003(All Units)	MCA03005(All Units)	MCA04001(All Units) MCA04002(All Units) MCA04003(All Units) MCA04004(All Electives) MCA04005(All Electives)		
1.2.1 New Course Introduced	Whole Program	Whole Program	Whole Program	Whole Program		
1.3.1 Cross-cutting issues relevant to Professional Ethics, Gender, Human Values, Environment and Sustainability						
1.3.2 Value-added Courses						
1.3.4 Field projects/ Research projects/ Internships	MCA01006	MCA02006	MCA03006	MCA0400		

Course Objectives:

1. Discuss about Computers and their Applications, as well as the fundamental concepts of computer hardware and software,
2. Explain the concept of CPU , Instruction Cycle , the concept of various number systems.
3. To discuss the concept of Systems Programming , Operating Systems , Language Translators.
4. Discuss the concept of Algorithm Development , Sorting and Searching and basic Data Structures .

UNIT-1

Computer Organization: Simple model of. Computer to explain how it executes algorithms, CPU, Memory, I/O Units: VDU, magnetic disk: floppy & hard disk, Optical Disk. Construction of drives and their specifications, Discussion of impact/non- impact types of printer, Graphics I/O devices.

UNIT-2

CPU Structure, Machine Instruction, Operation Codes and Operand Location, Instruction Counter, Fetch and Execute Cycles in a Computer, Arithmetic Operations, Memory Locations, Address and Contents, Read and write Operations. Data Representation: Integers, Real Numbers, Binary, Octal, Hexadecimal and their inter conversions, Binary Arithmetic.

UNIT-3

Elementary Systems Programming Concepts: Elementary idea of Operating Systems- Batch Processing, Multiprogramming, Time Sharing and real time systems. HLL& LLL. Language Translators: -Assemblers, Compilers, Interpreters. Difference between compiler and interpreter. Phases of Compilation.

UNIT-4

Algorithm Development: Problem Analysis, Flow Chart, Decision table. Fields, Vectors and files, Searching: linear & binary search. Hashing algorithms and resolution of collisions, typical hashing functions.

Internal Sorting Algorithms: discussion of algorithm with example for Straight selection sort, Bubble sort, Radix Sort, Quick sort, Heap sort, External Sorting Algorithms: Merge sort.

Basic Data Structures: Arrays, Linked Lists, Stack, Queue, Dequeue, Tree.

COURSE OUTCOME: Upon Completion of the course, students will be able to

CO1: Identify computer hardware and peripheral devices

CO2: Learn about Instruction Cycle and Differentiate between various number systems

CO3: Be Aware of the advantages and disadvantages of various Operating Systems, Computer Language and Language Translators.

CO4: Acquire proficiency in the writing of Algorithms. Construction of Flowcharts, and be familiar with different methods of Sorting, Searching, Data Structures.

Mapping of Course Content with Course Outcomes: MCA 01001

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Reading:

- Fundamentals of Computers –V Rajaraman Publisher- Prentice Hall India Learning Private Limited
- Publication date: 17 December 2014
- Computers and Commonsense- Hunt & Shelley Bibliographic information; Edition, 3, illustrated; Publisher, Prentice/Hall International, 1983
- An Introduction to Data Structures and Algorithms- Tremblay & Sorenson Publisher: McGraw-Hill, 1984
- Edition: 2
- Fundamentals of Computers- Reema Thareja- Publisher: OUP India Edition: Second, 2020

MCA

FIRST SEMESTER

Course Code: MCA 01002, Compulsory Theory Introduction to Digital Logic

Max. Marks: 100

L	T	P	C
6	0	0	6

Course Objectives:

The main objective of this course is to introduce the basic tools for the design of digital circuits and introducing methods and procedures suitable for a variety of digital design applications.

1. To make the students understand the digital system and Semi-Conductor Devices.
2. To understand the Boolean algebra and Logic Gates, K-Maps.
3. Knowing the building blocks of computer architecture.
4. Understanding the Arithmetic Operations and sequential circuit and microprocessors

UNIT -1

A Brief overview of Semiconductor Devices, Logic gates, Transistor – Transistor Logic, emitter- coupled logic, MOSFET logic, CMOS logic.

UNIT -2

Logic Components, Truth Table, Basic Gates, Design of Combinational circuits with AND, OR, NOT, NAND, NOR Gates, Boolean Algebra, K-Maps, Don't Care Condition in K-Maps.

UNIT -3

Flip-Flop, Registers, Counters, Adders Subtractors, Encoders and Decoders (with concept of MUX & DEMUX), ALU, Control, Memory and I/O.

UNIT-4

Introduction to Microprocessor: General structure of a Microprocessor- Timing and external control- Instruction set- Addressing Modes. Arithmetic operation with fixed point and floating-point data, Introduction to microprocessor Programming.

COURSE OUTCOME:

- CO1:** Gain both theoretical and experimental knowledge logic gates, Transistors and CMOS logics.
- CO2:** Learn about design various logic gates, and K-Map.
- CO3:** Understand computer architecture with the help of Flip-Flop and other components.
- CO4:** Learn about Microprocessor and Write programs using 8085 microprocessor

Mapping of Course Content with Course Outcomes: MCA 01002

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Reading:

- An Introduction to Digital Logic-Morris Mano, Publisher-Pearson India 2016.
- An Introduction to Microprocessors- Aditya Mathur, Publisher-Mc Graw Hills-1990
- Digital logic- M. Rafiquzzaman, Steven A. McNinch - Publisher: John Wiley & Sons Inc 2019
- Foundations of Computer Technology-Alexander John Anderson- Publisher Taylor & Francis Group 2020

MCA

FIRST SEMESTER

Course Code: MCA 01003, Compulsory Theory

Max. Marks: 100

Business Data Processing

L	T	P	C
6	0	0	6

Course Objectives

1. Learn about Information Systems in a Business Organization and Management Services.
2. Understand System analysis and feasibility study, system Flow Chart
3. Learn about System implementation, Documentation and design of System.
4. Understand the EPS, E-commerce Models, and Security Issues, cyber laws.

UNIT-1

Introduction to business organization: The business as a system, brief idea of various business subsystems: marketing, accounting, production, purchasing, personnel and inventory control. The role of a Management Services department.

UNIT-2

System analysis and design: The system life cycle, system planning- feasibility study, system investigation- establishing term of reference, fact finding and fact recording- the system flowchart.

UNIT-3

Physical design of the system: design of files, databases, input and output, procedures, design of dialogue, forms, codes. System implementation and documentation.

UNIT-4

Introduction, Definition, Objectives, Advantages and disadvantages of E-Commerce, Growth and Importance of E-Commerce.

E-Commerce Models: Business to consumer, Business to Business, Consumer to Consumer, other models.

Electronic Payment Systems: Special features required in payment systems, Types of E-payment systems, E-Cash, E-cheque, Credits card, Smart Card, Electronic Purses.

Security Issues in E-Commerce: Security risk of E-Commerce, Types of threats, Security tools. Cyber laws.

Course Outcomes: Students will be able to:

CO1: Recognize & appreciate the role of computing in a wide variety of activities & applications in modern society, including commerce, communication, education, travel & social interactions.

CO2: Understand System analysis and feasibility study, system Flow Chart.

CO3: Learn about System implementation, Documentation and design of System.

CO4: Articulate the implementation and relevance of the latest E-Commerce practices.

Mapping of Course Content with Course Outcomes: MCA 01003

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

- Data-Centric Business and Applications: -Publisher- Springer, by Natalia Kryvinska, Michal Greguš · 2020
- Bharat Bhaskar, Electronic Commerce – Framework Technologies and Applications, Publisher-Tata McGraw Hill , 2016.
- Ravi Kalakota & A.B. Whinston, Frontiers of Electronic Commerce, Publisher -Pearson Education , 2020

MCA

FIRST SEMESTER

Course Code: MCA 01004, Compulsory Theory

Max. Marks: 100

Design Methods and Analysis of Algorithms

L	T	P	C
6	0	0	6

COURSE OBJECTIVES:

1. To be able to analyze correctness and the running time of the basic algorithms for some classic problems in various domains and to be able to apply the algorithms and design techniques to advanced data structures, and to be able to analyze the running time complexity of different searching and sorting algorithms. To learn the divide and conquer technique.
2. To learn different types of algorithm design techniques like, dynamic programming, greedy method.
3. To learn backtracking, branch and bound and their applications in solving different practical problems.
4. To study different types of minimum cost spanning trees, shortest path algorithms, flow networks, string matching algorithms, NP-completeness and reductions.

UNIT-1

Simple Algorithms. Analyzing Algorithms, Asymptotic Notation, Recurrence relations. Design Methods: General Consideration, Algorithm design paradigms and representative Problems. Divide and Conquer (Binary search, Merge Sort, Quick Sort, Arithmetic with Large integers)

Unit – II

Greedy Method (Minimal Spanning Tree, Shortest Paths, Knapsack, etc.),
Dynamic Programming (Chained Matrix Multiplication, Optimal Storage on Tapes, Shortest Paths, Optimal Search Trees)

Unit – III

Backtracking (8-queens problem, Graph Colouring, Hamiltonian Cycles, etc.), Branch and Bound (0/1Knapsack problem, Travelling Salesperson), Approximation (Graph Colouring, Task Scheduling)

Unit – IV

Probabilistic Algorithms (Numerical Integration, Primality Testing). Graph Algorithms: BFS, DFS and its applications. Intractable Problems: Basic Concepts, Nondeterministic Algorithms, NP Completeness, Cook's Theorem, Examples of NP-Hard and NP-Complete problems. Problem Reduction.
Lower Bound Techniques.

COURSE OUTCOME: After the completion of the course, the students will be able to:

- CO1:** Know about different types of asymptotic notations that are used to analyze the running time of different algorithms and solve recurrences. Overview of Divide and Conquer paradigm with examples.
- CO2:** To know different types of algorithm design techniques like, dynamic programming, greedy method.
- CO3:** To know backtracking, branch and bound and their applications in solving different practical problems.
- CO4:** To know different types of minimum cost spanning trees, shortest path algorithms, flow networks, string matching algorithms, NP-completeness and reductions.

Mapping of Course Content with Course Outcomes: MCA 01004

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

1. A. Aho, V. Alfred, J. Hopcroft and J. D. Ullman, The Design and Analysis of Computer Algorithms, Addison Wesley. Pearson Publication-2006
2. E. Horowitz and S. Sahani, Fundamentals of Computer Algorithms, Galgotia, New Delhi. Publisher: Universities Press 2010
3. S.E. Goodman and S.T. Hedetniemi, Introduction to the Design and Analysis of Algorithms, McGraw Hill, 2007
4. G. Brassard and P. Bratley, Algorithmics, PHI, 2009

MCA

Course Code: MCA 01005, Compulsory Theory

PC Software Tools

FIRST SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

Course Objectives

The objective of the course is to introduce the concepts of basic MS-Office tools & their applications for the efficient use of office technology in a business environment.

UNIT-I

MS DOS: Basic Operating System concept, CUI, concept of files and directories, Booting of the system, Internal and External DOS commands, Partition of disk, Limitation of DOS.

MS Windows: Basic multiprogramming concept, GUI, Folders, Concept of login and logout, My Documents, My computer, My Network, Recycle Bin, Start Button, Task Bar, Date and Time setting, Calculator, Word Pad, System tools.

UNIT-II

MS Word: Opening, Creating, Saving a document, Editing, Finding and Replacing Text, Using the Interface (Toolbars and Menus).

MS-Word basics, Entering, Moving, Selecting and Copying text. Applying fonts and font styles, use of Numbering and Bullets. Finding and Replacing text. Spelling, Grammar and Thesaurus. Page formatting, print previewing and printing text. Creating headers and footers. Constructing tables in MS-Word. MS-Word Mail Merge, use of templates.

UNIT-III

MS Excel: Concept of Work book, Opening, Creating, Saving a work book and or organization of worksheets in a workbook, Data entry in cell, Selecting/Copying/Moving data in a worksheet. Introductory MS-Excel skills. Entering and Editing cell entries. Working with numbers. Changing the worksheet layout. Formatting Texts, Borders, and Colour, Printing in MS-Excel. Using functions and references. Naming ranges. Charting in MS-Excel. Creating Charts. Editing and Formatting Charts

UNIT-IV

Advanced Excel. Ms- Excel Financial and Statistical functions. Sorting and creating an MS- Excel database. Creating subtotals in MS-Excel. Using a data form. Creating Pivot tables. Using and creating MS-Excel templates. Linking Workbooks. Formulas between worksheets.

MS Power Point: Business presentation and their advantages, Opening, Creating, saving a presentation in MS-Power Point.

Course Outcomes

CO1: Student learn about Disk Operating System and concept of CUI & GUI.

CO2: They understand and learn how to use MS Word.

CO3: They understand and learn how to use MS Excel.

CO4: Demonstrate the technicalities of advanced Excel and creating a PowerPoint presentation & to Know the practices in data & file management.

Mapping of Course Content with Course Outcomes: MCA 01005

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings

- PC Software Made Simple: Taxali Publisher: McGraw Hill Education India-2001
- Sanjay Saxena: A First Course in Computers Vikas Publishing House -2003

MCA

Course Code: MCA 02001, Compulsory Theory

Programming in C

SECOND SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

Course Objectives

1. Learn how to solve common types of Problem Solving Techniques.
2. Learn data types and control structures of C
3. Learn to map problems to programming features of C such as Functions, Parameters Passing, Pointers, Arrays, Strings and Recursion.
4. Learn to write good portable C programs, with knowledge of C Preprocessors, C-Library, Unions, Input Output & file I/O.

UNIT-I

Basic Programming Concepts: Introduction to the basic ideas of problem solving and programming using principles of top-down modular design, Flowcharts, Abstraction Mechanisms, Stepwise Refinement. Syntactic Elements of a Language, General Syntactic Criterion, Formal Definition of Syntax, Semantics, Storage Management, Static & Dynamic.

UNIT-II

Introduction to Programming Language C: Data Types, Instruction and its Types, Storage Classes, Operators and Hierarchy of Operations, Expressions in C, Control and Repetitive Statements, If –Else, Else-if, Switch Loops while, For Do-While, Go to and labels, break, continue.

UNIT-III

Functions: User Defined Functions and Library Functions, Local and Global Variables, Parameter Passing, Pointers, Arrays, Strings, Recursion

UNIT-IV

C Preprocessors, Structures, Input and Output in C, C-Library. Unions, Input Output & file I/O. Introduction to the Major Programming Paradigms: Imperative Language, Object Oriented Languages, Functional Languages, Logic Languages, Parallel Languages etc.

Course Outcome (CO)

Upon successful completion of the course, a student will be able to:

CO1: Student able to understand programming concepts, mechanism, and syntax.

CO2: Analyze data types and control structures of C

CO3: Knowledge about C Functions, Parameters Passing, Pointers, Arrays, Strings and Recursion.

CO4: Understand concept of oops, parallel language with knowledge of C Preprocessors, C-Library, Unions, Input Output & file I/O.

Mapping of Course Content with Course Outcomes : MCA 02001

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

1. C Programming: Learn to Code- Sisir Kumar Jena · 2021, Taylor & Francis Ltd
2. Effective C by Robert C Seacord-2020, No Starch Press
3. C. Ghezzi & M. Jazayeri, Programming Languages Concepts, John Wiley-1996
4. B.W. Kernighan and D.M. Ritchie, the C Programming Language, PHI.1988
5. Pointer in C Programming by Thomas Mailund – 2021, A press

MCA
Course Code: MCA 02002, Compulsory Theory
Database Management Systems

SECOND SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVES: -

1. Acquire knowledge in fundamentals of Data Base Management System and be able to analyze the difference between traditional file system and DBMS, ER Modeling, data models.
2. Acquire knowledge of DDL, DML, Codd's Rule and Normalization Techniques
3. Able to write SQL queries mathematically and gain knowledge about how queries are being processed and executed.
4. Deal with online transactions and Concurrency control with knowledge of Data Base failures and Recovery, Basic knowledge of ORACLE.

Unit – I:

Data base concept: Data base design, design philosophies, DBMS Architecture, conceptual, logical and physical model, data base management systems, goals and objectives, classical data models-relational, hierarchical, network models. Database Users and Data Independence. ER Modeling,

Unit–II:

Detailed study of relational model: Data definition and data manipulation languages, relation types, role and Structural Constraints, Codd's Rule, application to data integrity and reliability, sharing concurrent access, consistency, security, privacy and authentication, Relational Database Design: Functional and multi-valued Dependencies, Desirable Properties of Decomposition, Normalization up to 5NF. Relational Algebra, Relational Calculus.

Unit – III:

SQL: Data definition and data types and literals, Specifying basic constraints, Schema change command, queries in SQL, Insert, Delete & Update commands, Types of SQL commands, SQL operators, Tables, views and indexes, Queries and sub queries, Aggregate functions, Cursors in SQL. Concept and Design of Object-Oriented Database.

Unit – IV:

Selected Database Issues: Security, Transaction Management, Basic Algorithms to Query Processing and Query Optimization, Concurrency Control, Recovery Techniques, Introduction to Oracle Architecture, Simple PL/SQL programming.

COURSE OUTCOMES (CO):

After the completion of the course, the student will be able to:

CO1: Learn about DBMS Architecture, ER-modelling and different Data Base languages and various data models for Data Base.

CO2: Students are understanding about the relational model, RDBMS design and Normalization, Acquire knowledge of DDL, DML, Codd's Rule and Normalization Techniques.

CO3: Learn about SQL and implication of SQL commands.

CO4: Deal with online transactions, control Concurrency and understand types of Data Base failures and Recovery with Basic knowledge of ORACLE.

Mapping of Course Content with Course Outcomes: MCA 02002

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

1. Data Base System Concepts, Abraham Silberschatz, Henry F. Korth, S. Sudarshan-2021, McGraw Hill.
3. Fundamental of Database Systems, Elmasri, Navathe- 2022, Pearson
4. Fundamental of Database Systems, Mark Gillenson – 2022, Wiley
5. R. Elmasri, and S.B. Navathe, Fundamentals of Database Systems, -1998.
6. Database Management Systems, McGraw-Hill Education.16 September 2002

MCA
Course Code: MCA 02003, Compulsory Theory
Web Technology and Information Retrieval

SECOND SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVES: After successful completion of this course, the students should be able to,

CO1: Understand the concepts of HTML, WWW Browsers, Web Publishing, Internet Basics.

CO2: Understand the concepts of Web Servers, Web Services. Firewalls. Internet Security, TCP/IP.

CO3: Understand the concepts of Indexing and Search: Boolean Queries and Inverted Index.

CO4: Understand the characteristics of Clustering and Classification, Social network analysis.

Unit-I

HTML, WWW Browsers, Web Publishing, Internet Basics, E-mail, File Transfer Protocol, Telnet, Usenet and Internet Relay Chat, Introduction to World Wide Web, Hypertext Data, Internet in India. Intranets.

Unit – II

Web Servers. Web Services. Firewalls. Internet Security, TCP/IP Address and URL's Domain Name Service. File Transfer Protocol. Telnet. Hypertext Transfer Protocol
Information Retrieval and Web Mining, Information Retrieval Concepts and Models, Search Engines, Crawling the Web.

Unit-III

Indexing and Search: Boolean Queries and Inverted Index, Relevance ranking, Similarity search, Web directories, Combining Searching with Browsing, Meta searchers, Web Query Languages, Dynamic Search and Software Agents.

Unit-IV

Clustering and Classification, Social network analysis, Measuring and Modeling the Web, Question answering, Semantic Web.

COURSE OUTCOME(CO): After successful completion of this course, the students should be able to,

CO1: Understand the concepts of WWW, browser, HTML, IRC, Web Publishing, Internet Basics.

CO2: Understand the concepts of Internet related things like- Web Servers, Web Services, firewall, FTP.

CO3: Understand the concepts of Indexing and Search, Boolean Queries and Inverted Index

CO4: Understand the characteristics of Clustering and classification, Semantic web, Social network analysis.

Mapping of Course Content with Course Outcomes: MCA 02003

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

1. Information Retrieval Technology-Fu Lee Wang, Haoran Xie, Wai Lam-2020, Springer
2. Advance in Information Retrieval by Leif Azzopardi, Benno Stein, Norbert Fuhr-2019, Springer
3. Evaluating Information Retrieval and Access Tasks by Tetsuya Sakai-2021

MCA

Course Code: MCA 02004, Compulsory Theory

Cyber Security & Cyber Laws

SECOND SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVE:

1. Understand the risks faced by Computer information systems and the nature of common Information hazards.
2. Maintain an appropriate level of awareness, knowledge and skill to develop Application Security with knowledge of security threats, Electronic Payment Systems.
3. Understand the threats and security mechanisms for Computer Networks, Information Security Governance & Risk Management, Security Architecture & Design.
4. To provide suitable coverage of security policies and cyber laws of India.

Unit – I

Introduction to information systems, Types of information systems, Development of information systems, Introduction to information security, Need for information Security, Threats to information Systems, Information Assurance, Cyber Security.

Unit-II

Application Security (Database, e-mail, Internet), Security Technology-Firewall and VPNs, Intrusion Detection, Access control. Security Threats- Viruses, Worms, Trojan Horse, Bombs, Trapdoors, Spoofs, E-mail Viruses, Macro viruses, malicious software, Network and Denial of services attack. Security Threats to E-commerce, Electronic Payment System, e-cash, Credits/Debit cards, Digital Signature.

Unit-III

Developing Secure Information System, Application Development security, Information Security Governance & Risk Management, Security Architecture & Design.

Unit-IV

Security policies, Salient Feature of IT Act 2000, Legal Provisions under the information Technology Act, Recent amendments by the IT (Amendment Act), 2008

COURSE OUTCOME: After the completion of the course, the student will be able to:

CO1: Understand the basics of information security and cyber laws.

CO2: Understand the various applications of information security, knowledge and skill to develop Application Security with knowledge of security threats, Electronic Payment Systems.

CO3: Develop secure information systems, security mechanisms for Computer Networks, Information Security Governance & Risk Management, Security Architecture & Design.

CO4: Understand the various cyber laws and acts followed for information security.

Mapping of Course Content with Course Outcomes: MCA 02004

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

- 1- Understanding Cyber Security Law and digital Privacy-Melissa Luking, Arash Habibi-2021, Springer
- 2- Cyber Crime and Information Technology- Alex Alexandrou- 2021, CRC Press
- 3- Cyber Law and Ethics-Mark Grabowski-2021, Pearson

MCA

**Course Code: MCA 02005, Compulsory Theory
System Programming**

SECOND SEMESTER

Max. Marks: 100

L	T	P	C
0	0	0	6

COURSE OBJECTIVE:

- Overview of systems programming, Components of system S/W.
- To understand of files, manuals, Directories, file properties and file systems.
- To know about Assembly Language, registers, addressing modes, instruction format, Op-code & mnemonics.
- To know about Evolution of Microprocessor, structure of micro processors, Introduction to simple Assembly language programming.

Unit- I

Components of system S/W, evolution of system S/W, general machine structure (Memory, Data, instructions), Machine language and assembly language.

Unit –II

Assembler: Need for assembler, design of assembler, one pass and two pass assembler, symbol table organization, working of a macro processor, loader, (compile and go, absolute, subroutine linkage relocating, direct linking).

Unit III

Assembly Language: Special purpose function of registers, concept of m-address machine, various addressing modes, instruction format, Op-code & mnemonics.

Unit IV

Different classes of Assembly Language Instructions: data transfer, arithmetic branch, loop, logical, shift and rotate instructions, I/O instructions, integer arithmetic. Introduction to simple Assembly language programming (AL of a simple microprocessor to be used as illustration). Evolution of Microprocessors, Introduction to Microprocessor Architecture.

COURSE OUTCOMES: At the end of this course students must be able to:

CO1: Overview of systems programming, Components of system S/W.

CO2 Understand files, manuals, Directories, file properties and file systems.

CO3 Understand the basics of Assembly Language and its Operations.

CO4: Understand the Development of Microprocessor and its basics Operations such as Arithmetic and logical.

Mapping of Course Content with Course Outcomes: MCA 02005

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

- Modern System Programming by Richard Whaling -2020
- System Programming by John J. Donovan- 2017, Mc.Graw Hill
- System Programming by D.M. Dhamdhare- 2021, Mc.Graw Hill
- Microcomputer System The 8086/8088 Family, Liu and Gibson, PHI.

MCA

Course Code: MCA 03001, Compulsory Theory

Operating System

THIRD SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVE:

1. To learn the fundamentals of Operating Systems, Operating system classification.
2. To know the components such as Schedulers and management aspects of concurrency management like Mutual exclusion algorithms, deadlock detection algorithms.
3. To learn the mechanisms involved in memory management in contemporary OS.
4. To learn Information management: File support, access methods, allocation methods, Device management.

Unit I

Introduction to OS: OS as resource manager, OS services, device management, Operating system classification: Single user, multiuser, simple monitor, batch processing, time sharing & real-time OS.

Unit II

Processor management-process states, multiprogramming, levels of schedulers and scheduling operations, deadlock detection, recovery & prevention, avoidance of deadlock.

Unit III

Memory Management: Partition, paging & segmentation, virtual memory and demand paging, procedure sharing, run time storage allocation.

Unit IV

Information management: File support, access methods, allocation methods-continuous, linked and indexed allocation, directory system, file protection, layered file system.

Device management, outline of a simple spooler, I/O programming. Exposure to systems like MS-DOS, UNIX, MS-WINDOWS etc.

COURSE OUTCOME:

After the completion of the course, the student will be able to:

CO1: Understand the role of operating system and explain the different structures of operating system.

CO2: Understand process management and compare the performance of various process scheduling algorithms.

CO3: Understand the concept of Memory Management.

CO4: Understand Information management: File support, access methods, allocation methods Information management and Device management.

Mapping of Course Content with Course Outcomes: MCA 03001

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Reading

1. Operating Systems – Dietel, 2023, Tata McGraw Hill.
2. Principles of Operating System by Sri V. Ramesh (2010), Pearson
3. Operating System Concepts – Silberschatz, Galvin, Gupta, Wiley, 2020
4. Operating System (Schaum Series)- Harris, 2022, Tata McGraw Hill.

MCA
Course Code: MCA 03002, Compulsory Theory
Discrete Mathematical Structures

THIRD SEMESTER			
Max. Marks: 100			
L	T	P	C
6	0	0	6

COURSE OBJECTIVE:

1. Apply the operations of sets and use Venn diagrams to solve applied problems; solve problems using the principle of inclusion-exclusion, to learn about the properties of Relations.
2. Determine the domain and range of a discrete or non-discrete function, graph functions, identify one-to-one functions, perform the composition of functions, find and/or graph the inverse of a function, and apply the properties of functions to application problems, to learn about the properties of Functions and Groups.
3. To learn about the properties of POSET and Lattice.
4. To learn about the properties of Boolean Algebras: Boolean Lattices, K-Maps.

Unit I

Basic concepts of set theory, Notation, Inclusion and equality of sets, Power set, Venn diagrams, basic set identities, ordered pairs and Cartesian product.

Relations: Relations, Properties of Binary relation in a set, Relation Matrix and graph of a relation, Equivalence relations, Binary Relation.

Unit II

Function: Definition and Introduction, Composition of a function, Inverse Function.

Groups: Algebraic system, Examples and definitions, Semigroup and Monoids, Groups, Subgroups, Homomorphism, Normal subgroups.

Unit III

Poset: Partial Ordering, Partially Ordered Set. Hasse diagram, Rules for drawing Hasse diagram and examples. Lattice: Lattice as poset, properties of lattice.

Unit IV

Boolean Algebras: Boolean Lattices and Boolean algebra, Uniqueness of finite Boolean algebra, Boolean function and Boolean Expressions, Minimization of a Boolean Expression using K-maps.

COURSE OUTCOMES:

At the end of the course the student will be able to:

CO1: Learn about Set theory and Relations

CO2: Learn the concept of Function and Group.

CO3: Understand the concept of POSET and Lattice.

CO4: Understand the concept of Boolean Algebra, and Boolean Function, Boolean Lattices, K-Maps.

Mapping of Course Content with Course Outcomes: MCA 03002

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Reading: -

- Discrete Mathematics and its Applications by Kenneth H Rosen 8th Edition-2020, McGraw Hill
- Discrete Mathematics: An Open Introduction by Oscar Levin-2018 Publisher, CreateSpace Independent Publishing Platform
- Discrete Mathematics and Applications by Kevin Ferland -2023, CRC Press

MCA

Course Code: MCA 03003, Compulsory Theory
Graph Theory & Combinatorics

THIRD SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

Course Objectives

1. To introduce the students to the concept of graphs, their properties and their applications.
2. To introduce Euler line and Euler Graphs, and Operations on Graphs and Trees, Spanning Trees.
3. To introduce Planar Graphs, Graph Coloring, Applications of Graph Theoretic.
4. To introduce Mathematical Logic Statements, Tautologies, Disjunctive and Conjunctive normal forms.

Unit – I

Graph theory: Introduction, Terminology and Application of Graphs, finite and infinite graphs, Isomorphism, Subgroups, walks path and circuit, connected and disconnected graph, Components of a graph

Unit – II

Euler line and Euler Graphs, Operations on Graphs, Transitive Closure, Hamiltonian paths and circuits, Shortest Path.

Trees: Properties of trees, distance and Centre in a tree, Rooted and Binary Trees, Spanning Trees

Unit–III

Planar Graphs: Definition, Matrix Representation: Incidence and Adjacency. Graph Coloring, Applications of Graph Theoretic Concepts to Computer Science.

Unit-IV

Mathematical logic Statement and Notations, Tautologies, Disjunctive and Conjunctive normal forms, equivalence of statements. Basic Combinatorics: Counting Techniques, Generating Functions, Recurrence Relations

COURSE OUTCOMES:

At the end of the course the student will be able to:

CO1: Learn about basics of Graph Theory; also learn about types and their components.

CO2: Learn about the concept of Euler Line, Euler Graphs, Graph Operations, Trees, Spanning Trees.

CO3: Learn about Planar Graphs and Matrix Representations, and understand how apply Graph concepts to Computer Science.

CO4: The students will be able to apply principles and concepts of Mathematical Logic, Generating Functions and Recurrence Relations.

Mapping of Course Content with Course Outcomes: MCA 03003

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

- Advance Graph Theory and Combinatorics by Michel Rigo-2016, Wiley
- A First Course in Graph Theory and Combinatorics: Second Edition 2022, Springer
- R. Johnsonbaugh, Discrete Mathematics, Pearson Education, 2001. Prentice Hall
- Fundamental of Graph Theory by Allen Bickle -2020, Springer

MCA

Course Code: MCA 03004, Compulsory Theory

Theory of Computation

THIRD SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVE: The learning objectives of this course are to:

1. Introduce students to the mathematical foundations of computation including automata theory; the theory of formal languages and grammars; the notions of algorithm, decidability, complexity, and computability.
2. Enhance/develop students' ability to understand Pumping Lemma, Context free languages, applications of Context-free grammars and languages, Chomsky and Greibach normal form.
3. Develop design push down automata and context free language, Closure property of CFL's.
4. Develop concept of Turing machine and able to design Turing machine for various languages.

Unit – I

Introduction to Automata and theory of Computation: alphabets, strings, languages, states, transitions, Deterministic and nondeterministic computations. Finite automata and regular expressions, applications of regular expressions: lexical analyzers and text editors

Unit – II

The pumping Lemma and closure property of regular sets. Context-free grammars, Chomsky and Greibach normal form, Ambiguity in context-free grammars, Context free languages, applications of Context-free grammars and languages.

Unit – III

Pushdown automata, Pumping Lemma for CFL's, Closure property of CFL's, Decision algorithms for CFL's. Relation of Languages to Automata

Unit-IV

Turing machines: Introduction, Hypothesis, Programming techniques, Restricted Turing machines, Turing machines and computers. Turing computable functions, Limitations Of Turing Machines.

COURSE OUTCOME: After the completion of the course, the student will be able to:

CO1. Learner will able to interpret mathematical Foundation of computation including automata theory.

CO2. Make use of Pumping Lemma, Context free languages, applications of Context-free grammars and languages, Chomsky and Greibach normal form.

CO3. Students will be able to understand and design push down automata and context free language, Closure property of CFL's.

CO4. Understand the concept of Turing machine and able to design Turing machine for various languages.

Mapping of Course Content with Course Outcomes: MCA 03004

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

1. Theory of Computation by D.P. Acharjya -2019, Springer
2. Introduction to Theory of Computation, M. Sipser, PWS Publishing Corporation, 1997.
3. Theory and Applications of models of Computations by Hubert Chan, Lap Chi Lau-2013
4. Introduction to Automata Theory, Languages and Computation- Hopcroft J.E., Ullman J.D. ADDISON-WESLEY, 2010.

MCA

Course Code: MCA 03005, Compulsory Theory
Object oriented Programming System with JAVA

THIRD SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVES:

1. To justify the philosophy of object-oriented design and the concepts of encapsulation, abstraction, inheritance and polymorphism.
2. To design simple programs in an object-oriented programming language using JAVA.
3. To design, implement and test the implementation of different types of inheritance relationships among objects and Interface and Package.
4. To compare and contrast the notions of Exception Handling , Multithreaded Programming , Applet Programming.

Unit -1

OOPS Concept: Object, Class, Inheritance, polymorphism, Message, Dynamic Binding, OOP Paradigm, Reusability and Extensibility, Encapsulation, Information Hiding, Genericity.

Unit – II

Introduction to Java: importance and feature of Java, Defining a Class, Adding a Variable and Method in Class, Creating Object, Constructor, Data types, Operators and expressions Decision making, branching and looping statements.

Array and Strings: Creating an Array, One and Two dimension Array, String Array and Methods, String and String Buffer Classes, Wrapper Classes

Unit – III

Inheritance and Polymorphism in Java: Types of Inheritance, Using Super (keyword), Final Variable and Method, Final Classes, Abstract Method and Classes, Method Overloading, Method Overriding, Visibility Control.

Interface and Package: Defining, Extending and implementing Interface, Accessing Interface variables, Java API Package, Naming Conventions, Creating Package, Accessing Package, Hiding Classes.

Unit – IV

Exception Handling: - Error, Types of errors, Exceptions, Syntax of Exception Handling code, Multiple Catch Statements, Using finally statements, Throwing own exception,

Multithreaded Programming: Fundamentals, Java Thread Models, Creating Threads, Extending the Threads, Stopping and Blocking a Thread, Life cycle of a Thread, Thread Priority.

Applet Programming: Local & Remote applet, Applet Life Cycle, Building Applet Code,

COURSE OUTCOME: After the completion of the course, the students will be able to:

CO1: Learn about Object Oriented Programming concepts like class, polymorphism etc.

CO2: Implement Methods, constructor, conditional statements and looping statements, array and string etc.

CO3: Implement inheritance efficiently and also study Interface and Package.

CO4: Learn about Exception Handling, Multithreaded Programming Object-Oriented Design and Programming using JAVA and Applet.

Mapping of Course Content with Course Outcomes: MCA 03005

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested reading: -

- Core Java for the Impatient by Cay S. Horstmann-2022, Pearson
- Murach'S Java Programming by Joel Murach-2022, McGraw Hill
- Let us Java by Kanetkar Yashavant-2019, Pearson
- Java the complete reference, 12th Edition by Herbert Schildt-2021 CRC Publication

MCA
Course Code: MCA 04001, Compulsory Theory
Software Engineering

FOURTH SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVES:

1. To learn the basic engineering principles and process models applicable for specific domains.
2. To understand the various estimation models and issues related to the software development projects.
3. To specify the important of design activities and methodologies in software product development, Planning Scheduling and Project Monitoring.
4. To able to understand the coding standards and testing mechanisms Implementation, Documentation & Maintenance.

UNIT-1

Introduction, Phases in Software Development, Software Development Process Models, Metrics.

UNIT-2

Software Requirements, Specification, Role of SRS, Structure Information, Data Flow Diagram & Data Dictionary, Structure Analysis & Design, Characteristics & Components of SRS, Requirements Document And Its Validation, Metrics For SRS Structuring Information, Data Flow Diagram & Data Dictionary, Structured Analysis & Design, Characteristics & Component Of SRS, Requirements Document & Validation, Metrics For SRS.

UNIT-3

Software Project Planning, Cost Estimation, Project Scheduling, Organization & Team Structure, Software Configuration Management, Project Monitoring & Controls, Technical Quality & Management Plans, Quality Assurance Plans.

UNIT-4

System Design, Detailed Design, Coding & Testing, Implementation, Documentation & Maintenance.

COURSE OUTCOME:

After the completion of the course, the student will be able to:

CO1: understand the concepts of software and engineering discipline in software development, process models and apply science, and engineering.

CO2: identify, formulate, and solve engineering problems, project planning and understanding characteristics of good software design.

CO3: understand the software quality models and use the techniques, skills, and modern engineering tools necessary for engineering practice.

CO4: understand the concepts re-engineering, reuse and maintenance of software product and able to develop SRS document and project Theory.

Mapping of Course Content with Course Outcomes: MCA 04001

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Reading

- Introduction to Software Engineering by Theodore Hammond · 2022, Pearson
- Software Engineering: Architecture-driven Software Development by Richard F Schmidt · 2013, CRC
- Introduction to Software Engineering by Ronald J. Leach · 2018, McGraw Hills
- Software Engineering for Absolute Beginners by Nico Loubser · 2021, Pearson
- An Integrated Approach to Software Engineering- Pankaj Jalote, Wiley-India, 2014.

MCA

Course Code: MCA 04002, Compulsory Theory

ARTIFICIAL INTELLIGENCE

FOURTH SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVES:

1. Understanding of introductory concepts and problem-solving techniques in Artificial Intelligence.
2. Getting insight to searching, Game Playing, AND-OR Graph.
3. Understand concept of Knowledge Representation, Introduction to Expert Systems
4. Getting concepts on Natural Language Processing, Pattern Recognition, Machine Learning.

UNIT-1

Artificial Intelligence: Definition, Historical Overview, Growth, Turing Test and Its Significance Branches of AI and Applications, Problem Solving, production system and Control Strategies.

UNIT-2

State Searching: Informed and Uninformed (blind) Searches-DFS, BFS, Best First Search etc. AND-OR Graph, Algorithms A*, Properties of A*, AO* and Related Algorithms, Game Playing Strategies, Mini- Max Procedure and Applications.

UNIT-3

Knowledge Representation: Propositional Logic, First Order Predicate, Semantic Net, Partitioned Semantic Net, Frames and Scripts, Conceptual Dependency (CD), Reasoning Process, Uncertainty Considerations.

Introduction to Expert Systems: System Feasibility Considerations, Architecture, Tools Overview of Rule Based and Other Types of ES Design.

UNIT-4

Natural Language Processing: Grammar for Natural Languages, Parsing, Transition Nets (TN), ATN and RTN Parser, Lexicon, Sentence Generation.

Introduction to: Pattern Recognition, Planning, Machine Translation, Neural Nets and Machine Learning, Fuzzy Logic.

COURSE OUTCOME:

CO1: Students will be able to assess critically the techniques presented and to apply them to real world problems. Get a introductory concepts and problem solving techniques in Artificial Intelligence.

CO2: Students will get to understand the major concepts of searching, Game Playing, AND-OR Graph.

CO3: Students will be able to understand concept of Knowledge Representation and Expert Systems in Artificial intelligence.

CO4: Student will be aware of Pattern Recognition, Machine Learning and Natural language processing in Artificial intelligence.

Mapping of Course Content with Course Outcomes: MCA 04002

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

- Artificial Intelligence for All: Transforming Every Aspect by Utpal Chakraborty · 2020 McGraw Hills
- Basics of Artificial Intelligence and Machine Learning by Dr Dheeraj Mehrotra · 2019 Pearson
- Russell & Norvig, Artificial Intelligence: A modern Approach, Pearson Education, 2013 Pearson
- Introduction to Artificial Intelligence by Wolfgang Ertel · 2018 Springer
- Artificial Intelligence – Rich and Knight, TMH, 2010

MCA
Course Code: MCA 04003, Compulsory Theory
Computer Communication Networks

FOURTH SEMESTER			
Max. Marks: 100			
L	T	P	C
6	0	0	6

COURSE OBJECTIVE:

1. Independently understand basic computer network technology, Data Communications System and its components, Measure of Information, Encoding of Source Output.
2. Performances Analysis and Compression, Error Detection and Correction Codes.
3. Identify Goals and Application of Networks, OSI reference model, networks topology, Physical Layer: Medium access sub layer.
4. Understanding the different networking layers: Network Layer, Transport Layer, Application Layer protocols.

UNIT-I

Digital Communication, Fundamentals of Digital Communication, Communication Channel, Measure of Information, Encoding of Source Output, Shannon's Algorithms, Discrete and Continuous Channel Entropy, Variable Length Code.

UNIT-II

Data Compression Shannon Hartley Theorem, Transmitter, Channel Noise, Amplitude Modulation, Frequency Modulation, Sampling, Pulse Modulation, PWM, PDM, PPM, PCM. Compression & Codes, Scheme Coherent and Non-Coherent Detector, Probability of Error, Performances Analysis and Compression, Error Detection and Correction Codes.

UNIT-III

Goals and Application of Networks, Networks Structure and architecture, the OSI reference model, services, networks topology, Physical Layer-transmission, switching methods, Integrated services digital networks, terminal handling.
Medium access sub layer: Channel allocations, LAN protocols, ALOHA Protocols-Pure ALOHA, slotted ALOHA, Carrier Sense Multiple Access Protocols, CSMA with Collision free Protocols, IEEE Standards, FDDI, Data Link Layer-elementary data link protocols, sliding windows protocols, error handling, high Level Data Link Control.

UNIT-IV

Network Layer: Point-to Point networks, routing algorithms, congestion control algorithms, Internetworking, TCP/IP packet and IP addresses, IPv6.

Transport Layer: Design issues, connection management, TCP window Management, User Datagram Protocol, Transmission Control Protocol.

Application Layer: Network Security DES, RSA algorithms, Domain Name System, Simple Network Management Protocol, Electronic mail, File Transfer Protocol, Hyper Text Transfer Protocol, Cryptography and Compression Techniques.

COURSE OUTCOME: After the completion of the course, the student will be able to:

CO1: Understand the computer network technology, Data Communications System and its components, Measure of Information, Encoding of Source Output.

CO2: Understand the Analysis and Compression, Error Detection and Correction Codes.

CO3: Understand the application of Networks, OSI reference model, networks topology, Physical Layer: Medium access sub layer.

CO4: Understand the operation of different networking layers: Network Layer, Transport Layer, Application Layer protocols.

Mapping of Course Content with Course Outcomes: MCA 04003

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

- Computer and Communication Networks by Nader F. Mir · 2014
- Data Communication and Computer Networks by Dr M. P. Vani, Notion Press; 1st edition, 2019
- Computing in Communication Networks: From Theory to Practice by Frank H.P. Fitzek, Fabrizio Granelli, Patrick Seeling · 2020 Springer

MCA

Course Code: MCA 04004A, (Elective Theory)

Computer Graphics

FOURTH SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVES:

1. Understand the Scan Conversion of various geometrical forms and Region filling and acquire Knowledge about the Two-Dimensional Transformations
2. Two-Dimensional Viewing and Clipping of lines and polygons
3. Understand the Three-Dimensional Graphics and concept of Projection
4. To be familiar with solid area scan conversion and different types of curves and surface design.

UNIT-I

Introduction and Line Generation: Types of computer graphics, graphic displays- random scan displays, raster scan displays, frame buffer and video controller, points and lines, line drawing algorithms, circle generating algorithms, midpoint circle generating algorithm, and parallel version of these algorithms.

UNIT-II

Transformations: Basic transformation, matrix representations and homogenous coordinates, composite transformations, reflections and shearing.

Windowing and Clipping: Viewing pipeline, viewing transformations, 2-D clipping algorithms- Line clipping algorithms such as Liang Barsky algorithm, polygon clipping, curve clipping, and text clipping.

UNIT-III

Three Dimensional: 3-D geometric primitives, 3-D object representation, 3-D transformation, 3-D viewing, projections, and 3-D clipping.

UNIT-IV

Curves and Surfaces: Quadric surfaces, spheres, ellipsoid. Hidden lines and surfaces: back face detection algorithm, depth buffer method, a- buffer method, scan line method, basic illumination models– ambient light, diffuse reflection, warn model, intensity attenuation, color consideration, transparency and shadows.

COURSE OUTCOME:

After the completion of the course, the student will be able to:

CO1: - Students will get the Practical concepts of Graphics display devices, different types of graphics drawing algorithms.

CO2: - Students will get the concepts how to use of 2D and 3D Geometrical Transformations.

CO3: - Students will get the practical concepts of Hidden Line/surface elimination techniques.

CO4: - Students will get the idea of Multimedia concepts and also the concept of Computer Animation

Mapping of Course Content with Course Outcomes: MCA 04004

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

- Computer Graphics from Scratch by Gabriel Gambetta · 2021 Springer
- Computer Graphics with An Introduction to Multimedia by Chopra Rajiv · 2017 CRC
- Fundamentals of Computer Graphics by Steve Marschner, Peter Shirley · 2018 McGraw Hill

MCA

Course Code: MCA 04004B, (Elective Theory)

Digital Image Processing

FOURTH SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVES:

1. The fundamentals of digital image processing
2. Image transform used in digital image processing
3. Image enhancement techniques used in digital image processing
4. Image restoration techniques and methods used in digital image processing and Image compression and Segmentation used in digital image processing

UNIT-I

Why Digital Images; The Digital Camera; Data Types And 2d Representation of Digital Images; Discrete Sampling Model; Quantization; Noise Processes; Image Attributes Thresholding and Thresholding Algorithms; Performance Evaluation And ROC Analysis; Connected Components Labeling; Region Growing And Region Adjacency Graph (RAG); Split And Merge Algorithms; Grey Level Transformations; Histogram Equalization; Geometric Transformations; Affine Transformation; Polynomial Warps.

UNIT-II

Erode And Dilate As Max And Min Operators On Binary Images; Open, Close, Thinning And Other Transforms; Medial Axis Transform; Introduction To Grey- Level Morphology; Calculation Of Region Properties; Moment Features; Boundary Coding; Fourier Descriptors Line Descriptors From Boundary Coding And From Moments.

UNIT-III

Linear And Non-Linear Filtering Operations; Image Convolutions; Separable Convolutions Sub- Sampling And Interpolation As Convolution Operations; Alternative Approaches; Edge Enhancement By Differentiation; Effect of Noise, Edge Detection And Canny; Implementation; Edge Detector Performance Evaluation, Image Structure Tensor; Relationship to Image Auto- Correlation; Characterization And Harris Corner Detector.

UNIT-IV

Sub-Pixel Accuracy And Performance Evaluation; Representations of Colour In Digital Images; Colour Metrics; Pixel- Wise (Point) Operations; Colour Invariants And Finlayson Colour Constancy Algorithm similarity and Dissimilarity Matching Metrics; L2 Metric And Relationship To Cross- Correlation; Image Search And Multi- Resolution Algorithms, 2D Object Detection, Recognition, Location

COURSE OUTCOME: After the completion of the course, the student will be able to:

CO1: To discuss on the basics of digital image processing and digital image formation. To illustrate different mathematical preliminaries to deal with digital image processing.

CO2: To be able to Develop Fourier transform for image processing in frequency domain and image Enhancement: Spatial Domain

CO3: To be able to evaluate the methodologies for image segmentation, restoration, enhancement

CO4: Understand the need for image compression and multi resolution analysis and to learn the spatial and frequency domain techniques of image compression.

Mapping of Course Content with Course Outcomes: MCA 04004B

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Reading: -

- Digital Image Processing: Practical Approach by Borko Furht, Esad Akar, Whitney Angelica Andrews · 2018
- Introduction to Digital Image Processing by William K. Pratt · 2013
- Understanding Digital Image Processing by Vipin Tyagi · 2021

MCA

Course Code: MCA 04004C, (Elective Theory)

Machine Learning

FOURTH SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVE: The learning objectives of this course are to:

1. Understand basics Machine Learning algorithms and their applications.
2. Understand modern notions in Supervised Machine Learning and its concepts- Regression, Clustering, Support Vector Machine.
3. Understand and use of Neural Network , Back Propagation Learning , Recurrent Neural Networks, Deep Learning techniques.
4. Understand and implement Genetic Algorithms.

Unit I

Introduction of Machine Learning, Basics of Linear Algebra and Statistics, Overview of target function representations; Supervised and unsupervised learning

Unit II

Regression-Least Mean Square and Recursive Least Square Algorithms; Clustering: K-Means, Hierarchical, and Density-based Clustering, Spectral Clustering. Decision Tree, Overfitting and Pruning, Logistic regression, Support Vector Machine and Kernel; Noise, bias-variance trade- off, under-fitting and over-fitting concepts.

Unit III

Single Layer Neural Network, Multilayer Perceptron, Back Propagation Learning, Functional Link Artificial Neural Network, and Radial Basis Function Network, Recurrent Neural Networks, Deep Learning, Convolutional Neural Network.

Unit IV

Genetic Algorithm, Schemata Theorem, Differential Evolution, Particle Swarm Optimization, Ant Colony Optimization, Convergence Analysis.

COURSE OUTCOME:

CO1: Student will be capable to understand basics of Machine learning including Supervised Learning, Unsupervised Learning, Ensemble Learning, Reinforcement Learning concepts.

CO2: Students will be capable to understand Supervised Learning techniques and use them in real life problem.

CO3: Student may be capable to get the concept and use of Neural Network, Back Propagation Learning, Recurrent Neural Networks, Deep Learning techniques.

CO4: Students will be introduced with advanced implement Genetic Algorithms for various applications.

Mapping of Course Content with Course Outcomes: MCA 04004C

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Reading:

- Machine Learning for Absolute Beginners by Oliver Theobald · 2017, Scatterplot Press
- Machine Learning and Its Application by Indranath Chatterjee · 2021 CRC
- Fundamentals of Machine Learning for Predictive Data by John D. Kelleher, Brian Mac Namee, Aoife D'Arcy · 2020

MCA

Course Code: MCA 04005A, (Elective Theory)

Soft Computing

FOURTH SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVE:

1. To know Neural Network techniques and its applications.
2. Get knowledge of fuzzy logic. To know applications of advanced Neural Networks.
3. Get knowledge of Advanced topics in fuzzy logic.
4. To know various Fuzzy image processing, Fuzzy Data Fusion; Fuzzy Diagnosis, Neural Networks, GA.

UNIT-I

Neural Networks: History, overview of biological Neuro-system, mathematical Models of Neurons, ANN architecture, Learning rules, Learning Paradigms- Supervised, Unsupervised and reinforcement Learning, ANN training Algorithms-perceptions, Training rules, Delta, Back Propagation Algorithm, Multilayer Perception Model, Hopfield Networks, Associative Memories, Applications of Artificial Neural Networks.

UNIT-II

Introduction to Fuzzy Logic, Classical and Fuzzy Sets: Overview of Classical Sets, Membership Function, Fuzzy rule generation on Fuzzy Sets: Compliment, Intersection, Unions, Combinations of Operations, and Aggregation Operations.

UNIT-III

Fuzzy Numbers, Linguistic Variables, Arithmetic Operations on Intervals & Numbers, Lattice of Fuzzy Numbers, Fuzzy Equations. Fuzzy Logic: Classical Logic, Multi valued Logics, Fuzzy Propositions, Fuzzy Qualifiers, Linguistic Hedges. Uncertainty based Information: Information & Uncertainty, Non specificity of Fuzzy & Crisp Sets, and Fuzziness of Fuzzy Sets.

UNIT-IV

Introduction of Fuzzy image processing, Fuzzy Data Fusion; Fuzzy Diagnosis; Neural Networks: Supervised Learning: Hopfield Nets, Perceptron, gradient descent, multilayer nets, back propagation, over fitting. Introduction of Neuro-Fuzzy Systems, Architecture of Neuro Fuzzy Networks. Application of Fuzzy Logic: Medicine, Economics etc. An Overview, GA in problem solving, Implementation of GA

COURSE OUTCOME: After the completion of the course, the student will be able to:

CO1: understand various Soft Computing basic and evolution of computing.

CO2: understand about Genetic algorithm and optimization techniques.

CO3: understand Neural Network techniques and its applications.

CO4: understand about fuzzy logic and its applications.

Mapping of Course Content with Course Outcomes: MCA 04005A

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

- Applied Soft Computing: Techniques and Applications by Samarjeet Borah, Ranjit Panigrahi · 2022, Apple Academic
- Advanced Soft Computing Techniques in Data Science by Sujata, Subhendu, Pani, Ajith · 2021
- Soft Computing by Samir Roy, Udit Chakraborty · 2013, Springer
- Concepts of Soft Computing: Fuzzy and ANN with Programming by Snehashish Chakraverty, Deepti Moya Sahoo, Nisha Rani Mahato · 2019, Springer

MCA
Course Code: MCA 04005B, (Elective Theory)
Data Warehousing and Data Mining

FOURTH SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVE:

1. To understand the principles of Data warehousing and Data Mining.
2. To be familiar with the Data warehouse architecture and its Implementation.
3. To know the Architecture of a Data Mining system.
4. To understand the various Data pre-processing Methods.
5. To perform classification and prediction of data.

UNIT-I

Data Warehousing and Business Analysis: - Data warehousing Components –Building a Data warehouse –Data Warehouse Architecture – DBMS Schemas for Decision Support – Data Extraction, Clean-up, and Transformation Tools –Metadata – reporting – Query tools and Applications – Online Analytical Processing (OLAP) – OLAP and Multidimensional Data Analysis.

UNIT-II

Data Mining: - Data Mining Functionalities – Data Pre-processing – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization and Concept Hierarchy Generation- Architecture Of A Typical Data Mining Systems- Classification Of Data Mining Systems.

Association Rule Mining: - Efficient and Scalable Frequent Item set Mining Methods – Mining Various Kinds of Association Rules – Association Mining to Correlation Analysis – Constraint-Based Association Mining.

UNIT-III

Classification and Prediction: - Issues Regarding Classification and Prediction – Classification by Decision Tree Introduction – Bayesian Classification – Rule Based Classification – Classification by Back propagation – Support Vector Machines – Associative Classification – Lazy Learners – Other Classification Methods – Prediction – Accuracy and Error Measures – Evaluating the Accuracy of a Classifier or Predictor – Ensemble Methods – Model Section.

UNIT-IV

Cluster Analysis: - Types of Data in Cluster Analysis – A Categorization of Major Clustering Methods – Partitioning Methods – Hierarchical methods – Density-Based Methods – Grid-Based Methods – Model-Based Clustering Methods – Clustering High-Dimensional Data – Constraint-Based Cluster Analysis – Outlier Analysis.

Mining Object, Spatial, Multimedia, Text and Web Data:

Multidimensional Analysis and Descriptive Mining of Complex Data Objects – Spatial Data Mining – Multimedia Data Mining – Text Mining – Mining the World Wide Web.

COURSE OUTCOME: After the completion of the course, the student will be able to:

CO1: To gain understanding of the principles of Data warehousing and Data Mining.

CO2: Possess knowledge of Data warehouse architecture and its Implementation

CO3: To know the Architecture of a Data Mining system and the the various Data preprocessing Methods..

CO4: Possess Technical knowhow of the Data Mining principles and techniques for real time applications.

Mapping of Course Content with Course Outcomes: MCA 04005B

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

1. Jiawei Han, Micheline Kamber and Jian Pei “Data Mining Concepts and Techniques”, Third Edition, Elsevier, 2011.
2. Alex Berson and Stephen J. Smith “Data Warehousing, Data Mining & OLAP”, Tata McGraw – Hill Edition, Tenth Reprint 2007.
3. K.P. Soman, Shyam Diwakar and V. Ajay “Insight into Data mining Theory and Practice”, Easter Economy Edition, Prentice Hall of India, 2006.
4. G. K. Gupta “Introduction to Data Mining with Case Studies”, Easter Economy Edition, Prentice Hall of India, 2006.
5. Pang-Ning Tan, Michael Steinbach and Vipin Kumar “Introduction to Data Mining”, Pearson Education, 2007.

MCA

Course Code: MCA 04005C, (Elective Theory)

Software Project Management

FOURTH SEMESTER

Max. Marks: 100

L	T	P	C
6	0	0	6

COURSE OBJECTIVE:

1. To know Fundamentals of Software Project Management (SPM).
2. Get knowledge of Project Organization and Scheduling.
3. Get knowledge of Project Monitoring and Control.
4. To know various aspects of Software Quality Assurance and Testing, Software Quality and CASE Tools.

UNIT-I

Introduction and Software Project Planning Fundamentals of Software Project Management (SPM), Need Identification, Vision and Scope document, Project Management Cycle, SPM Objectives, Management Spectrum, SPM Framework, Software Project Planning, Planning Objectives, Project Plan, Types of project plan, Structure of a Software Project Management Plan, Software project estimation, Estimation methods, Estimation models, Decision process.

UNIT-II

Project Organization and Scheduling Project Elements, Work Breakdown Structure (WBS), Types of WBS, Functions, Activities and Tasks, Project Life Cycle and Product Life Cycle, Ways to Organize Personnel, Project schedule, Scheduling Objectives, Building the project schedule, Scheduling terminology and techniques, Network Diagrams: PERT, CPM, Bar Charts: Milestone Charts, Gantt Charts.

UNIT-III

Project Monitoring and Control Dimensions of Project Monitoring & Control, Earned Value Analysis, Earned Value Indicators: Budgeted Cost for Work Scheduled (BCWS), Cost Variance (CV), Schedule Variance (SV), Cost Performance Index (CPI), Schedule Performance Index (SPI), Interpretation of Earned Value Indicators, Error Tracking, Software Reviews, Types of Review: Inspections, Desk checks, Walkthroughs, Code Reviews, Pair Programming.

UNIT-IV

Software Quality Assurance and Testing , Testing Objectives, Testing Principles, Test Plans, Test Cases, Types of Testing, Levels of Testing, Test Strategies, Program Correctness, Program Verification & validation, Testing Automation & Testing Tools, Concept of Software Quality, Software Quality Attributes, Software Quality Metrics and Indicators, The SEI Capability Maturity Model CMM), SQA Activities, Formal SQA Approaches: Proof of correctness, Statistical quality assurance, Cleanroom process.

Project Management and Project Management Tools Software Configuration Management, Risk Management: Risks and risk types, Risk Management Process, Software Project Management Tools: CASE Tools, Planning and Scheduling Tools, MS-Project.

COURSE OUTCOME: After the completion of the course, the student will be able to:

- CO1:** To know Fundamentals of Software Project Management (SPM).
- CO2:** Get knowledge of Project Organization and Scheduling.
- CO3:** Get knowledge of Project Monitoring and Control.
- CO4:** To know various aspects of Software Quality Assurance and Testing, Software Quality and CASE Tools.

Mapping of Course Content with Course Outcomes: MCA 04005C

Course Outcomes (Numbers)	CO1	CO2	CO3	CO4
Mapping of Course Content with Course Outcome	Unit I	Unit II	Unit III	Unit IV

Suggested Readings:

Software Project Management by M. Cotterell

Software Project Management by S. A. Kelkar